An A.M.O.R.C. Note on Quasi-Infinite Graph Color Bags

Lean-verified graph constructions and a four-color search envelope

Abstract

We formalize a conservative reading of the phrase "quasi-infinite adjoint" for simple graphs. The useful construction is not a reflector into colorable graphs. Instead, it is an arbitrary indexed coproduct of graph factors, with countable edge unions and inverse-limit thread graphs as complementary infinite constructions. The Lean development proves the coproduct universal property, propagation of heterogeneous colorings through a sigma-sum palette, a universal property for countable edge unions, inherited colorings for inverse limits, and a small obstruction theorem showing why a naive free colorable reflector fails. It then generalizes coloring to selected template pullbacks and simultaneous multicolor constraints.

The addendum connects this machinery to quantum spatial search: the four-color theorem gives every finite planar graph a complete multipartite four-color completion. That completion is Laplacian integral, so the deterministic-search theorem of Li, Luo, Feng, and Li applies to the completion, though not automatically to the original planar graph. More generally, equal-fiber joins over engineered regular templates inherit the template spectral ratio relevant to Grover-optimal CTQW search.

Formal Setting

A simple graph on a vertex type V is represented by an adjacency predicate $\mathrm{Adj} : V \to V \to \mathrm{Prop}$ together with symmetry and irreflexivity. A graph homomorphism $G \to H$ is a vertex function preserving adjacency. A coloring by a color type C is a homomorphism into the complete graph K_C, where $x \ \mathrm{Adj} \ y$ means $x \neq y$.

The Lean file uses a local `Bijection` structure rather than importing a full category theory library. This keeps the artifact small while still stating the relevant adjoint-shaped equivalences explicitly.

The Bag Construction

Given an index type I, vertex families V_i, and graphs G_i on V_i, define

$$\mathrm{vertices}(\mathrm{SigmaGraph}(G)) = \sum_{i \in I} V_i.$$

Two tagged vertices are adjacent only when they have the same tag and the underlying vertices are adjacent in that factor:

$$(i, a) \ \mathrm{Adj} \ (j, b) \Leftrightarrow \exists k, i = k \wedge j = k \wedge a \mathrm{Adj}_{G_k} b.$$

Lean verifies the universal property:

$$\mathrm{Hom}\left(\sum_{i \in I} G_i, H\right) \sim= \prod_{i \in I} \mathrm{Hom}(G_i, H).$$

This is the most literal successful version of the "bag of factors" idea. It is arbitrary in the index type, so it can be finite, countable, or larger, depending on the ambient universe.

Heterogeneous Color Factors

If every factor has its own palette C_i and coloring

$$\mathrm{color}_i : G_i \to K_{C_i},$$

then the whole bag is colored by the sigma-sum palette:

$$\sum_{i \in I} G_i \to K_{\sum_{i \in I} C_i}, \qquad (i, a) \mapsto \left(i, \mathrm{color}_{i(a)}\right).$$

If all factors share one palette C, the coproduct reuses that same palette. This captures the heterogeneous case from the motivating conversation without forcing all components to have the same chromatic number.

Countable Approximation

For graphs G_n on a shared vertex type V, Lean defines an edge-union graph:

$$x \text{ Adj } y \Leftrightarrow \exists n, x\mathrm{Adj}_{G_n} y.$$

A homomorphism out of this union is equivalent to one vertex map that preserves adjacency at every stage:

$$\mathrm{Hom}(\cup_n G_n, H) \sim= \left\{f : V \to W \mid \forall n, f \quad \text{preserves} \quad \mathrm{Adj}_{G_n}\right\}.$$

This is a lightweight colimit-style theorem for staged approximants. It is a reasonable formal proxy for “take the finite constructions and let them run quasi-infinitely”, while still avoiding unverified analytic claims about metric or fractal dimension.

Inverse Threads

For an inverse system of vertex types

$$\ldots \to V_2 \to V_1 \to V_0,$$

Lean defines a thread as a sequence $x_n \in V_n$ compatible with the bonding maps. The inverse-limit graph makes two threads adjacent when all finite coordinates are adjacent. Each coordinate projection is a graph homomorphism:

$$\pi_n : \lim G_n \to G_n.$$

Consequently, any coloring of one finite coordinate induces a coloring of the inverse-limit graph by composition with π_n.

The Failed Reflector

One tempting reading of “quasi-infinite adjoint” is: construct a free or largest C-colorable graph associated to any graph G. In the ordinary graph homomorphism category, that version fails before it gets started.

Lean proves:

$$\neg\ \mathrm{Colorable}(G, C) \wedge \mathrm{Colorable}(H, C) \Rightarrow \neg\exists(G \to H).$$

The reason is direct: if $G \to H$ and $H \to K_C$, then composition gives $G \to K_C$, contradicting non-colorability of G.

The development also formalizes the concrete witness: the triangle is not two-colorable, so there is no homomorphism from the triangle into any Bool-colorable graph. Thus an ordinary reflector with a unit $G \to L(G)$ into the full subcategory of two-colorable graphs cannot exist.

The Better Adjoint-Shaped Object

The successful adjoint-shaped object is local over a chosen vertex-color map. Fix $c : V \to C$. Among all graph structures on V for which c is a coloring, there is a greatest one:

$$K_{c(x,y)} \Leftrightarrow c(x) \neq c(y).$$

In other words, K_c is the cofree or maximal graph compatible with the color map. If $G \to K_C$ is a coloring, then G is a spanning subgraph of this completion. Lean verifies this as `ColorCompletion.greatest`.

For a genuine bag $V = \sum_{i \in I} V_i$, the map $\mathrm{tag}(i, a) = i$ gives

$$\mathrm{CompleteJoin}(V_i) = K_{\mathrm{tag}}.$$

This graph has all cross-bag edges and no within-bag edges. Lean verifies:

- `CompleteJoin.no_intra`: vertices in the same bag are never adjacent;
- `CompleteJoin.cross`: vertices in different bags are always adjacent;
- `CompleteJoin.greatest`: any graph colored by the tag projection is a spanning subgraph of the complete join.

So the quasi-infinite construction is not merely "take a bag". It is:

$$\text{colored graph} \to \text{independent color fibers} \to \text{complete cross-fiber host}.$$

The host is canonical once the coloring is fixed, and the index set can be arbitrary. For four-color planar graphs the index set is bounded by four; for the earlier speculative infinite construction, the fibers may be infinite while the color quotient remains finite.

Engineered Colorability

The better generalization is to stop treating a palette as a complete graph. Let Q be any graph on a type T of engineered modes, and let $f : V \to T$ be a selected label map. Define

$$f^*Q(x, y) \Leftrightarrow Q(f(x), f(y)).$$

Lean calls this `PullbackGraph Q f`. It is the greatest graph on V for which the chosen label map is a homomorphism into Q. Ordinary color completion is the special case $Q = K_C$.

For bags, this becomes an engineered join. Given fibers V_i over template vertices $i \in I$,

$$\mathrm{TemplateJoin}(Q, V)((i, a), (j, b)) \Leftrightarrow Q(i, j).$$

So a small template graph programs which bags are completely coupled. The old `CompleteJoin` is the special case where Q is complete; a bipartite template, cycle template, hypercube template, expander template, or Laplacian-integral template gives a different engineered host. Lean verifies the no-intra-fiber rule, the cross-fiber rule, and maximality as `TemplateJoin.greatest`.

Multicolorability is the intersection version. Given many templates Q_a and many selected label maps $f_a : V \to T_a$, Lean defines `MultiPullbackGraph` by

$$x \text{ Adj } y \Leftrightarrow \forall a, Q_{a\left(f_{a(x)}, f_{a(y)}\right)}.$$

This captures simultaneous constraints: planar color class, hardware zone, allowed long-range tunneling family, parity, oracle class, or any other selected coarse observable. The design problem changes from "is the graph colorable?" to "which quotient templates do we want the Hamiltonian support to factor through?"

A.M.O.R.C. Addendum: Four-Color Search Envelope

The fetched spatial-search papers make the relevant graph invariant clear. In the Chakraborty–Novo–Ambainis–Omar CTQW model, the search Hamiltonian is

$$H_G = -P_w - \gamma A_G,$$

where A_G is the adjacency matrix. Their optimality criterion is spectral: the largest adjacency eigenvector should be the uniform state and the rest of the normalized spectrum should be bounded away from it. King et al. use a Laplacian search Hamiltonian

$$H_\alpha = -\gamma_0 L_\alpha - P_w,$$

and identify the Laplacian spectral gap/algebraic connectivity as the critical resource for long-range tunneling search. Li–Luo–Feng–Li prove that any Laplacian integral graph admits deterministic search with certainty when the marked proportion is known in advance. Sadowski studies spatial search directly on planar Apollonian networks and finds that planar sparsity alone is not enough to make all marked-location cases operationally uniform.

This suggests a modest but real theorem.

Theorem (four-color Laplacian-integral envelope). Let $G = (V, E)$ be a finite loopless planar graph. Choose a four-coloring $c : V \to \{1, 2, 3, 4\}$, supplied by the four-color theorem. Define the completion K_c on the same vertex set by

$$x \mathrm{Adj}_{K_c} y \Leftrightarrow c(x) \neq c(y).$$

Then:

- G is a spanning subgraph of K_c;
- K_c is complete multipartite with at most four parts;
- K_c is Laplacian integral;
- therefore the deterministic quantum search theorem for Laplacian integral graphs applies to K_c.

Proof. If $x \mathrm{Adj}_G y$, then properness of c gives $c(x) \neq c(y)$, so the identity on vertices is a graph homomorphism $G \to K_c$. The definition of K_c connects exactly the vertices in distinct color classes, hence K_c is complete multipartite.

Let the nonempty color classes have sizes $n_1, ..., n_r$, with $r \leq 4$ and $N = n_1 + ... + n_r$. For the complete multipartite graph $K_{n_1,...,n_r}$, a vector summing to zero inside part i is a Laplacian eigenvector with eigenvalue $N - n_i$, giving multiplicity $n_i - 1$. On the subspace of vectors constant on each part, the Laplacian acts by $a_i \mapsto N a_i - \sum_j n_j a_j$, whose eigenvalues are 0 once and N with multiplicity $r - 1$. Thus all Laplacian eigenvalues are integers:

$$\mathrm{spec}(L(K_c)) = \{0, N, N - n_1, ..., N - n_r\}.$$

So K_c is Laplacian integral. Li–Luo–Feng–Li's deterministic-search theorem applies to every Laplacian integral graph, hence to K_c.

The Lean artifact verifies the graph-theoretic part of this theorem: `ColorCompletion` constructs K_c, `Coloring.toColorCompletion` proves the identity homomorphism $G \to K_c$, and the same-color/different-color lemmas prove the complete multipartite edge rule. The Laplacian spectrum calculation is ordinary linear algebra and is not yet mechanized in this repository.

Hamiltonian Reading

After ordering vertices by color classes, the adjacency matrix of a planar graph has zero diagonal color blocks:

$$A_G = \begin{pmatrix} 0 & A_{12} & A_{13} & A_{14} \\ A_{21} & 0 & A_{23} & A_{24} \\ A_{31} & A_{32} & 0 & A_{34} \\ A_{41} & A_{42} & A_{43} & 0 \end{pmatrix}.$$

The four-color completion replaces each off-diagonal block A_{ij} by an all-ones rectangular block. In CTQW language, it is a graph-engineering move: the planar Hamiltonian support is embedded into a four-sector long-range Hamiltonian support with complete cross-color tunneling. This connects the four-color theorem to spatial search without claiming that the original planar graph is Grover-optimal or deterministic-search-ready.

There is a stronger reduction hidden here. Let the color classes have sizes $n_1, ..., n_r$, $r \leq 4$, and let e_v denote the standard basis state at vertex v. Define

$$u_i = \frac{1}{\sqrt{n_i}} \sum_{v \in V_i} e_v$$

as the normalized uniform state on color class i. The span of these r states is invariant under both the adjacency and Laplacian of K_c. In this basis,

$$A_{K_c} u_j = \sum_{i \neq j} \sqrt{n_i n_j} u_i.$$

Thus the unmarked walk on the complete color host collapses from N dimensions to the weighted complete graph on the color bags.

The same calculation works for an engineered template join. If $J = \mathrm{TemplateJoin}(Q, V)$, then

$$A_J u_j = \sum_{i : Q(i,j)} \sqrt{n_i n_j} u_i.$$

So the uniform-fiber sector is not merely small; it is the weighted adjacency matrix of the chosen template. If all fibers have the same size m, then this sector is exactly mA_Q. For a d-regular template Q, the full adjacency spectrum of the equal-fiber join consists of m times the adjacency spectrum of Q, together with additional zero modes internal to the fibers. Therefore, if Q has uniform principal eigenvector and all other adjacency eigenvalues satisfy $|\lambda_i| \leq cd$ for some $c < 1$, then the equal-fiber join satisfies the same spectral-ratio condition used by Chakraborty–Novo–Ambainis–Omar for Grover-optimal CTQW spatial search. The bag construction therefore gives a family of arbitrarily large search hosts controlled by a small engineered template.

There is a parallel deterministic-search statement. If the fibers all have size m and the template Q is Laplacian integral, then the equal-fiber template join is Laplacian integral: the quotient Laplacian contributes m times the Laplacian spectrum of Q, and the internal fiber-difference modes contribute integer eigenvalues md_i. Thus Li–Luo–Feng–Li's deterministic-search theorem applies to equal-fiber joins over Laplacian-integral templates. Complete multipartite four-color completions are a special, more forgiving case where the fiber sizes need not be equal.

For search with a single marked vertex $w \in V_m$, refine the bag partition to

$$\{w\}, \quad V_m - \{w\}, \quad V_i \ \text{ for } \ i \neq m.$$

This refined partition is equitable for K_c. Consequently the search Hamiltonians

$$-\gamma A_{K_c} - P_w \quad \text{and} \quad -\gamma L_{K_c} - P_w$$

preserve the span of the normalized cell states. Since $r \leq 4$, a single-marked search on the four-color completion reduces to a Hamiltonian of dimension at most five. With an arbitrary marked set M, splitting each color bag into $V_i \cap M$ and $V_i - M$ gives an invariant subspace of dimension at most eight.

This is the more operational quantum link: four-coloring gives a bounded-sector Hamiltonian reduction. The graph may have arbitrarily many vertices, or arise as a countable/coherent limit of finite planar approximants, while the color-sector quotient remains bounded by four before marking and by eight after marking. That is the part that deserves the name quasi-infinite: the vertex space can grow without changing the color-sector control problem.

The same observation gives a boundary condition for the long-range tunneling paper: if all nonzero long-range couplings are treated as edges, then the support graph is complete for $N > 4$, hence not planar and not four-colorable. So four-color structure is a finite-range or thresholded-support invariant; it does not survive an all-positive long-range completion except as a chosen decomposition of the original planar skeleton.

What Was Actually Obtained

The verified construction is aggressive in size but conservative in claims:

- arbitrary indexed coproducts of graph factors;
- heterogeneous and shared-palette color propagation;
- a countable edge-union universal property;
- inverse-limit thread graphs with projections and inherited colorings;
- a formal obstruction to the naive colorable-reflector idea;
- a four-color completion theorem connecting planar graphs to Laplacian integral deterministic-search host graphs;
- a maximal complete-join construction over color bags;
- engineered template joins and multicolor pullback constraints;
- a spectral-ratio inheritance theorem for equal-fiber template joins;
- a bounded-dimensional equitable quotient for search Hamiltonians on the four-color completion.

This gives a precise mathematical home for a large part of the phrase "quasi-infinite adjoint": it is the coproduct/colimit/inverse-limit toolkit for colorable graph components. It does not yet construct graphs with a designed fractal dimension. To do that honestly, the next layer would need a metric or growth structure on the approximants, then a verified dimension invariant.

The A.M.O.R.C. addendum gives a concrete bridge to quantum spatial search: four-colorability provides canonical Laplacian-integral completions for planar instances, and those completions have bounded-dimensional color-sector search quotients. The bridge is structural, not an optimality theorem for the original graph.

Lean Artifact

The formalization is in:

```
Graphplay/Basic.lean
```

It builds with Lean 4.30.0 using:

```
lake build
```

References Fetched

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